



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (E&C) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EC701/20EC701/17EC703 - POWER ELECTRONICS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	With neat diagrams, explain the V-I characteristics of a thyristor.	10	L*2	1	1
b)	With neat circuit diagram and waveforms, explain the ON-OFF type of AC-voltage controllers. Derive equation for RMS output voltage.	10	L2	2	1
2. a)	Explain the two-transistor model of a thyristor.	8	L2	1	1
b)	Briefly explain the turn-on methods of thyristor.	6	L2	1	1
c)	Discuss the dynamic turn-off characteristics of a thyristor.	6	L2	1	1
3. a)	With neat circuit and waveforms explain the 1- ϕ full converter operation with inductive load. Derive equations for average and RMS output voltage.	10	L2	2	1
b)	With neat circuit diagram and waveforms, explain the working of a single-phase full wave AC Voltage controller. Derive equation for RMS output voltage.	10	L2	2	1
Unit – II					
4. a)	Discuss the turn-on base drive control technique for a BJT.	6	L2	3	1
b)	With neat circuit diagram and waveforms, explain buck regulator. Discuss the different modes of operation.	6	L2	4	1
c)	With neat waveforms, explain the switching characteristics of a power MOSFET.	8	L2	3	1
5. a)	The β of a bipolar transistor varies from 12 to 75. The load resistance is $R_C=1.5 \Omega$. The dc supply voltage is $V_{CC}=40 \text{ V}$ and the input voltage to the base circuit is $V_B=6 \text{ V}$. If $V_{CE(sat)}=1.2 \text{ V}$ and $V_{BE(sat)}=1.6 \text{ V}$, and $R_B=0.7 \Omega$, determine				
i)	The overdrive factor, (ODF)				
ii)	The forced β_f				
iii)	Power loss in transistor, P_T	8	L3	3	2
b)	With neat circuit diagram and waveforms, explain buck-boost regulator. Discuss the different modes of operation.	6	L2	4	1
c)	Illustrate the Comparison of Power Transistors.	6	L2	3	1
6. a)	With neat diagrams and waveforms, explain the switching characteristics of a power BJT.	8	L2	3	1
b)	With neat circuit diagram and waveforms, explain boost regulator. Discuss the different modes of operation.	6	L2	4	1
c)	Discuss the turn-on gate drive control technique for a MOSFET.	6	L2	3	1

P.T.O.

Unit – III

7. a) Briefly explain the performance parameters of inverters.
 b) With neat circuit diagram, explain variable dc link inverter
 c) Explain the working principle of a single-phase bridge inverter connected to a resistive load. Derive an expression for the output RMS voltage.
8. a) A single-phase full-bridge inverter as in Fig.Q.8.(a) has a resistive load of $R = 2.4 \Omega$ and the dc input voltage is ($V_s = 48 \text{ V}$). Determine the following;
 Rms output voltage at the fundamental frequency
 Output power
 Average and peak currents of each transistor

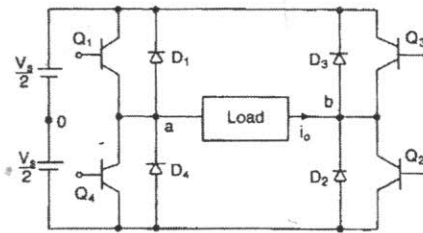


Figure.Q.8.(a)

- b) With neat circuit diagram and waveforms, explain transistor based current source inverter.
 c) With neat diagrams, explain the single-pulse-width modulation technique.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
